



भारतीय विज्ञान शिक्षा एवं अनुसंधान संस्थान मोहाली

(मानव संसाधन विकास मंत्रालय का एक स्वायत्त संस्थान, भारत सरकार के अधीन)
सेक्टर-81, नॉलेज सिटी, पो.ओ. मनौली, एस.ए.एस. नगर, मोहाली, पंजाब-140306

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MOHALI**

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NAME:

ROLL NUMBER:

MID-SEMESTER EXAM EES301

Date: 01.08.2025

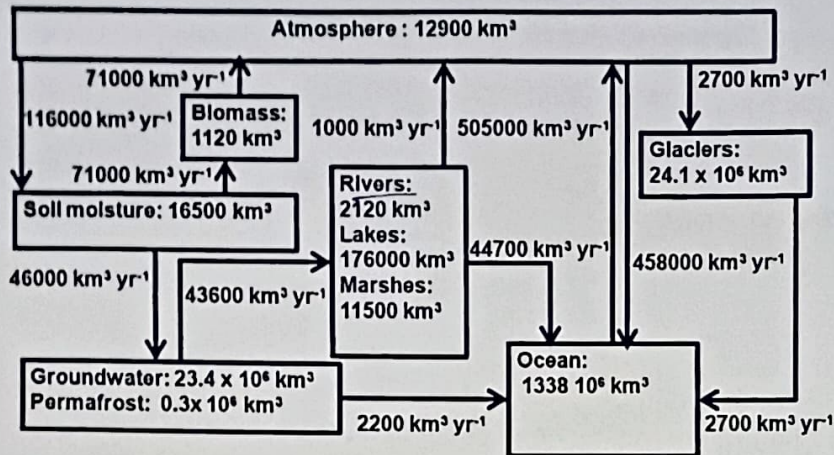
Global biogeochemical cycles and climate change

Total Marks: 25

Please read the questions carefully before answering and return the question paper with your name and roll number.

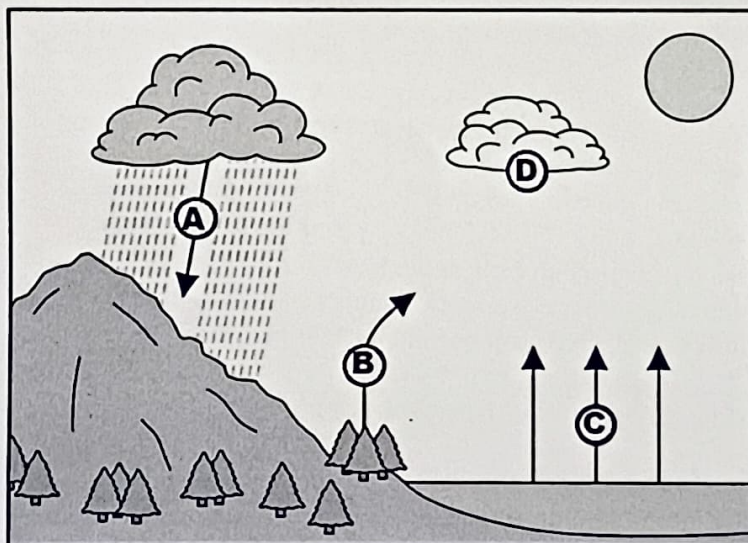
- 1) What do you need to do to test the hypothesis "acid rain damages plants"? (1 Mark)
 - a) Look up the credibility of the person or source proposing the hypothesis
 - b) Frame a testable null hypothesis "acid rain has no impact on plants" and your alternate hypothesis "acid rain damages plants", and design a suitable experiment to evaluate the null hypothesis
 - c) Design an experiment that could prove the hypothesis "acid rain damages plants" to be true
 - d) Design an experiment that could potentially prove the hypothesis "acid rain damages plants" to be wrong
- 2) Scientists want to perform an experiment to see if ozone pollution lowers moong dal yield. How should the control group in this experiment be treated? (1 Mark)
 - a) The control group should have moong dal plants growing at higher oxygen levels.
 - b) The control group should have moong dal plants growing in ozone free air.
 - c) The control group should have moong dal plants growing in different amounts of water.
 - d) The control group should have moong dal plants growing in ozone polluted air.
- 3) The early Universe was quite uniform. What then were the Goldilocks Conditions for the formation of stars? (1 Mark)
 - a) Tiny variations in the density of matter
 - b) Large variations in temperature
 - c) The existence of gravity
 - d) A concentration of heavy metals
- 4) What percentage of the Earth is made up of elements other than Hydrogen and Helium? (1 Mark)
 - a) 45%
 - b) 2%
 - c) 90%
 - d) 85%
- 5) Why was the first mini threshold of life's evolution, photosynthesis, so important? (1 Mark)
 - a) It allowed life to spread beyond oceanic vents.
 - b) It helped plants move onto land.
 - c) It transformed our atmosphere from oxygen-rich to carbon dioxide-rich.
 - d) It forced organisms to become multicellular to process more complex energy.
- 6) Calculate the turnover time and turnover rate of a reservoir containing 450×10^{12} kg carbon with a sink flux of 25×10^{12} kg y^{-1} (1 Mark)
- 7) If the turnover rate of a reservoir is 10% then what is the average turnover time / residence time of molecules in the reservoir (1 Mark).
- 8) An air quality station recorded the following 24hr average value in ambient air: air temperature= 305 K; air pressure = 1 atm; $R = 0.082$ L atm mol^{-1} K^{-1} ; $NO_2 = 70$ ppb. Convert to microgram per m^3 (M.W. of $NO_2 = 46$ g/mol) (2 Marks)
- 9) Evaluate the alkalinities of 0.2 M K_2CO_3 (1 Mark)
- 10) How many gram of NH_4Cl do you need to put into 1 L of solution to make 0.1 M NH_4Cl solution? $M_N=14$, $M_H=1$, $M_{Cl}=35.45$, $N_A=6.022 \times 10^{23}$. (1 Mark)
- 11) A first-order reaction is found to have a rate constant, $k=4.0 \times 10^{-3} s^{-1}$. What is the half life of the substance in question. (1 Mark)

12) If 100 g water contains 2 mg of uranium what is the concentration of uranium in ppm.



13) Calculate the turnover time of the sum of the surface water (Rivers+Lakes+Marshes) reservoir (1 Mark). Is the reservoir in steady state or not? (1 Mark)

- 14) Give the general definition of a “positive feedback cycle” and a “negative feedback cycle” using the terms effector & stimulus. (1 Mark)
- 15) Sketch the sea ice positive feedback cycle and explain how it can accelerate climate change and result in a non-vegetated Earth system flipping between a Snowball Earth and hothouse state. (1 Mark)
- 16) Sketch the accelerated silicate weathering climate feedback cycle and explain how it worked as a negative feedback which counteracted the increase in solar intensity with time. (1 Mark)
- 17) Explain the difference between elements and isotopes (1 Mark)
- 18) How did Earth lose its first H & He atmosphere and when (1 Mark)?
- 19) Where did Earth’s second atmosphere come from and what gases was it composed off (1 Mark)?
- 20) When and how did Earth’s third atmosphere start to form (1 Mark)?
- 21) Which abiotic substances critical to the evolution of life may have been delivered by meteorites and what is the key differences between the biotic and abiotic synthesis of these compounds? (1 Mark)



22) Name the processes A, B, C and D (2 Marks)